

Ambient Systems

AI-Driven Compliance & Sustainability Solutions for the Built Environment

Company Overview

Ambient Systems is an AI-driven compliance and sustainability venture re-establishing its Singapore operations to pilot collaborations under BEAMP Cycle 6. Originally spun out of the National University of Singapore in 2021, we've evolved into a global innovation hub with strategic operations across three continents.

- **(HQ)** – Core operations and system architecture
- **Hong Kong** – Asia-Pacific market validation and R&D
- **Québec** – North American research partnerships

Recognition & Patents

WIPO Patent

WO – AI Compliance System for construction verification

Panasonic Winner

Deep Tech Challenge 2023 – Innovation in built environment solutions

Google Cloud

-Up 2022 – AI deployment excellence

Business Activities

We develop AI systems for compliance automation, ESG intelligence, and performance verification across the construction and infrastructure sectors. Our solutions bridge the gap between regulatory requirements, operational excellence, and sustainability reporting.

Construction QA/QC

Automated quality assurance and control systems for building projects, ensuring compliance and performance standards are consistently met.

Infrastructure Digital Twins

Real-time virtual replicas of physical assets enabling predictive maintenance, optimization, and lifecycle management.

Sustainability Analytics

ESG reporting and green finance intelligence tools that track carbon metrics and support environmental compliance.

BEAMP Relevance & Singapore Focus

Built Environment Localization

Adapting our platform for Singapore's regulatory framework and construction standards.

Integrated Innovation

Combining FinTech, GreenTech, and ConTech modules for comprehensive solutions.

Automated Verification

AI-powered ESG reporting and compliance validation at scale.

Global Market Experience

1 **2021 – Singapore/ASEAN**

Early pilots and academic collaborations through NUS

2 **2023 – Japan**

Smart City Conference presentations at City Tech Tokyo

3 **2023 – Hong Kong/China**

R&D expansion and market validation initiatives

4 **2024 – North America**

Research partnerships and pilot deployments

5 **2024 – Europe**

Market expansion and technology refinement

Products & Services

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BEAMP Cycle 6 POC/Pilot Focus

Our proof-of-concept initiatives target three critical areas where AI can transform construction compliance and sustainability.

- **Concrete Quality** – Automated verification of delivery documentation, slump tests, and cube-test data with real-time alerts
- **Construction Economics** – QA performance metrics linked directly to financial milestones and project cost visibility
- **Sustainability Metrics** – Embodied carbon tracking and materials documentation for ESG reporting requirements

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Platform – ambientsystems.ai

A comprehensive digital ecosystem integrating compliance, analytics, and operational intelligence for the built environment.

- **Financial Metrics Integration** – Connect quality assurance data with cost tracking and budget performance
- **Generative AI Planning** – Scenario modeling for compliance workflows and logistics optimization
- **IoT Connectivity** – Seamless sensor integration with automated data capture and validation

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Consulting – ivandamnjanovic.com

Specialized advisory services leveraging our patented technology and deep domain expertise.

- Civil Engineering performance audits and compliance reviews
- Energy systems optimization and carbon footprint analytics
- Controlled Environment Agriculture (CEA) digital twin development
- Custom IP and system design based on WO patent

Business Targets

BEAMP Cycle 6 represents a strategic inflection point for Ambient Systems—serving as both the anchor for our Singapore-based innovation hub and the launchpad for global deployment across construction, finance, and ESG markets.

Establish Singapore Entity

Launch new legal structure focused on AI compliance and ESG solutions tailored for the built environment sector.

Transfer IP Portfolio

Register complete intellectual property portfolio including WO patent under Singapore operations.

Complete Platform Modules

Finalize development of Ambient Verify and Ambient Twin platform components for commercial deployment.

Secure Pilot Partners

Execute LOIs and MOUs with pilot clients and ecosystem partners across construction and finance sectors.

Advance Technology Readiness

Progress from TRL 6 to TRL 9 through successful BEAMP pilot implementations and real-world validation.

: 2025–2028

2025 Q1-Q2	2025 Q3-Q4	2026-2027	2028
R&D Phase	**Pilot Deployment**	**Commercial Scale**	**Global Deployment**
Platform refinement, IP transfer, pilot recruitment	BEAMP implementations, data collection, validation	Market expansion, partnership development	Regional hubs, enterprise clients

Key Team Members

Ivan Damnjanovic

Founder & Systems Architect

Ivan brings a decade of experience in technical management, risk analysis, and AI deployment, with over five years working across Asia-Pacific markets including Singapore, Nepal, and Hong Kong.

He has led R&D projects under prestigious institutions including NUS, NSERC, and Centennial College, and holds PhD-level expertise in Business Management & Economics.

Patent holder (WO) and lead architect of both Ambient Verify and Ambient Twin platforms.

Languages: Serbian (native), English (fluent), French (intermediate)

David Damnjanovic

Advertising & Communications

David serves as our creative specialist, bringing extensive experience in digital advertising and brand communications through his work at Yourbow.com.

He leads campaign strategy and execution for our online ventures, ensuring our message resonates with stakeholders across construction, finance, and technology sectors.

His expertise in visual storytelling and digital marketing strengthens our market positioning and stakeholder engagement initiatives.

Planned Singapore Hire (2025)

Full-Stack Developer / Data Engineer – We're committed to local talent development and will onboard a Singapore-based engineer to support pilot deployment, platform localization, and ongoing innovation initiatives under BEAMP Cycle 6.